

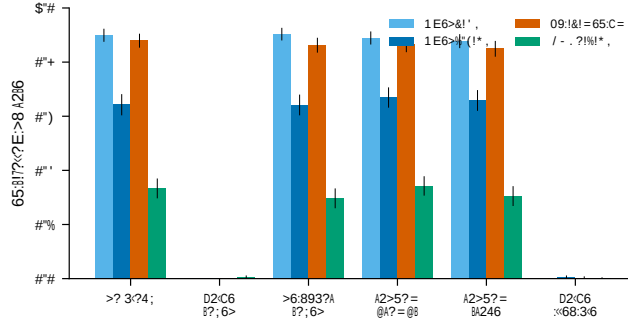


Figure 2: Rate at which the answer follows the edited value under each blocked attention target. Blocking the value’s own position collapses following on every model; matched control blocks do not. Error bars are 95 percent bootstrap intervals.

Retrieval occurs in the middle third of the network. Layer-resolved versions of both interventions locate the read in depth on Qwen3-4B (36 layers). Overwriting the value’s hidden state at any single layer from 0 through 22 redirects the answer at 0.90 to 0.94, matching the full multi-layer patch (0.92), while single-layer patches at layer 24 and beyond redirect at 0.02 or less ($n = 60$): the register can be rewritten anywhere upstream of its consumers and nowhere downstream of them. Blocking the value-position attention only at the middle third of layers (12 through 23) collapses following from 0.857 to 0.025, blocking only the early or late third leaves it unchanged (0.857 and 0.863), and no single layer’s block matters (0.84 to 0.89 at every layer, $n = 80$): the read is distributed redundantly across the middle third’s layers and confined to it. The two sweeps agree on the same boundary, near layer 23: the value is written into the position’s residual stream from early depth, retrieved by attention across the middle third, and causally inert afterward. Coarse sweeps on two further models (appendix) show the structure generalizes while the boundary’s depth is model-specific: on Phi-3-medium, early- and mid-third patches redirect at 0.89 and 0.87 and late-third patches at 0.007, matching the Qwen3-4B shape, while on Qwen2.5-7B patches in every third redirect at 0.70 to 0.73, so its reads extend into the final third of its 28 layers.

Reading is the preferred path, not the only one. Replacing the value’s characters with unreadable dots, without touching attention, also removes following, and differently by model: the Qwen models reconstruct the value from the previous line and recover the correct answer in 0.60 to 0.73 of cases, while Phi-3-medium does not recover (0.00).

6 When the written value is read versus recomputed

Models read the register by default. On DeepSeek V3.2, an edited value the prompt merely implies is followed at 0.93, while an edited value the prompt generatively defines (the prompt states the two numbers whose sum becomes the running value) is reverted at 0.43; with the defining sentence present but the edit two steps away, following returns to 0.96, so the reversion targets the defined value. Claude Sonnet 4.5 recomputes more broadly, reverting most when a stated checkpoint is the value’s only source. Llama 3.3 70B follows edits at 0.97 whether or not the value is prompt-defined, despite solving the same chains from scratch, so the capacity to recompute does not imply recomputing. Qwen models at 4B and 7B never revert. These conditions come from separate experiments; each within-experiment contrast is controlled, and we claim neither a single ordered scale across models nor that reconstruction cost governs the choice, which remains open.

The same test bounds the mechanism on natural benchmarks and bears on a disputed question. On GSM8K worked solutions, an edited intermediate is followed at 0.87 by Qwen3-4B and at 0.10 by V3.2, since a benchmark intermediate is often one operation from numbers still in the prompt and a stronger model recomputes it. Within one model, V3.2 follows edits at 0.93 on chains it cannot regenerate and at 0.10 on GSM8K. Lanham et al. [2023] report reliance on written reasoning declining with scale on standard benchmarks, a result later work disputes; our contrasts indicate the direction depends on the state itself. On regenerable state, stronger models depend less on the trace;

on state no model can regenerate, dependence is high at every scale we test. Faithfulness of the trace to the computation is a property of the state’s regenerability, and a single scale trend should not be expected.

7 Related work

Our interventions build on causal tracing and activation patching [Meng et al., 2022] and causal abstraction [Geiger et al., 2021]; Heimersheim and Nanda [2024] caution that broad patches conflate a component mattering with deleting information at a position, which the single-layer sweep in Section 5 addresses. The memory role of written tokens in prior work is discussed in the introduction [Merrill and Sabharwal, 2024, Lanchantin et al., 2023, Zhu et al., 2025, Shih et al., 2026]. Faithfulness studies [Turpin et al., 2023, Lanham et al., 2023] and filler-token results [Pfau et al., 2024] bound our claims to state the model does not regenerate internally.

8 Limitations

Tasks are numeric and count state tracking, so our claims concern serial task state rather than working memory in general; code, plans, and multi-entity state are untested. The J-space result bounds one fitted lens at generation-preserving doses and does not measure the J-space content of the written positions themselves. The fine layer sweeps are established on one model; the cross-model sweeps are coarse, and on Qwen2.5-7B they do not resolve the read boundary. White-box evidence is at 4B to 14B; frontier evidence is behavioral. The read-versus-recompute results support within-experiment contrasts, and the only patch control is size-matched noise; a structured control planting a valid activation for a different quantity would further isolate value content.

9 Conclusion

Serial reasoning state outgrows a language model’s internal workspace within one to two dependent steps at every scale tested, writing begins at no threshold, and no intervention shifts state between the workspace and the trace, so the chain of thought is working memory rather than overflow. It works as compositional registers: settable by intervention, retrieved through attention in the middle layers, independently addressable, and decisive for the answer. For oversight the implication cuts both ways: the written positions are a causally used memory channel, and identical text can carry different stored values, so trace monitoring observes the channel while verifying the state behind it remains a separate problem.

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